

AI Failures We Found

Assignment 4 — Smart Water Lab

Mahmudul Hasan (4125999049)

2026

1 Master table

AI output	Why wrong	How detected	Fix
S=25400*CN-254	Uses $\times$ not $\div$	Terminal Q $\approx$ 0.002 vs 43.6 mm	Division form in <code>scs_cn.py</code>
“P=5, CN=95 $\Rightarrow$ Q=0”	Ia $\approx$ 2.67 mm	Demo script AssertionError	Regression test
Runoff without Q $\leq$ P cap	Water balance	Parametrized pytest	<code>min(Q,P)</code>
Missing API rain field	Crashes	50-city pytest	Return 0 mm/h
SLSQP infeasible start	Storage bounds	<code>validate_scheduler</code>	rust-constr
Flood depth no floor	Negative depth	Physical check-list	<code>max(0,.)</code>

2 Statistics

52 AI outputs reviewed; **9** corrections; capstone **33** pytest; experiments **88** pytest; total **121**.

3 Evidence

Case B: `assignment3/evidence/buggy_rainfall.py`. Case A: `assignment3/scripts/demo_wrong_assumption`

4 Repository

<https://github.com/mahmud456alhasan-debug/smart-water-capstone>